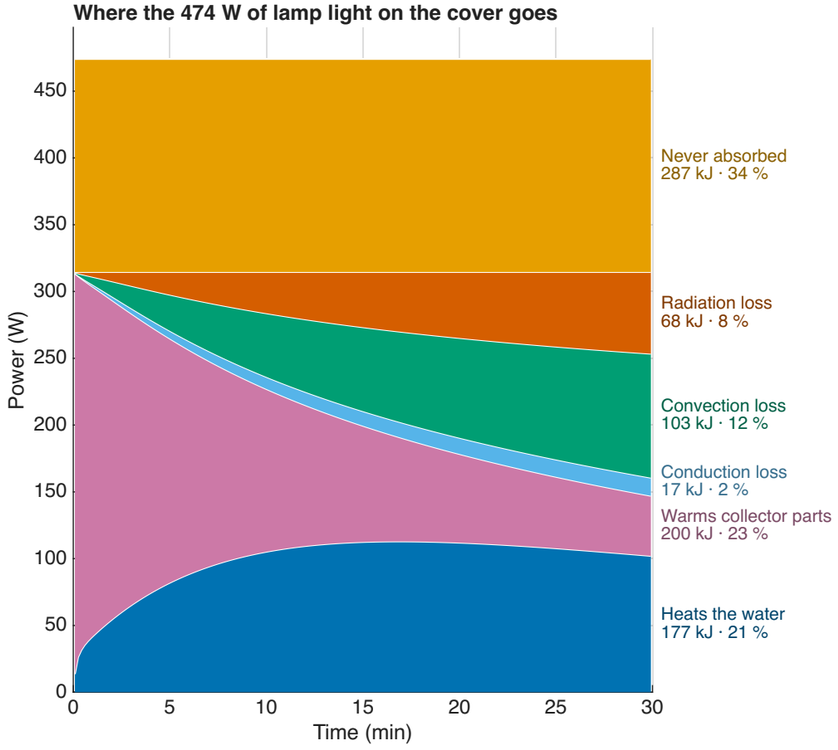
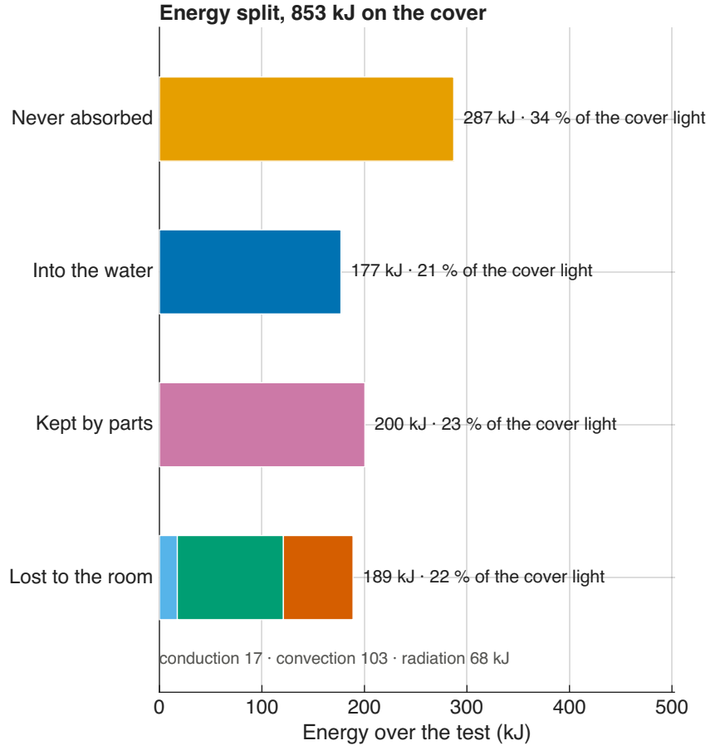
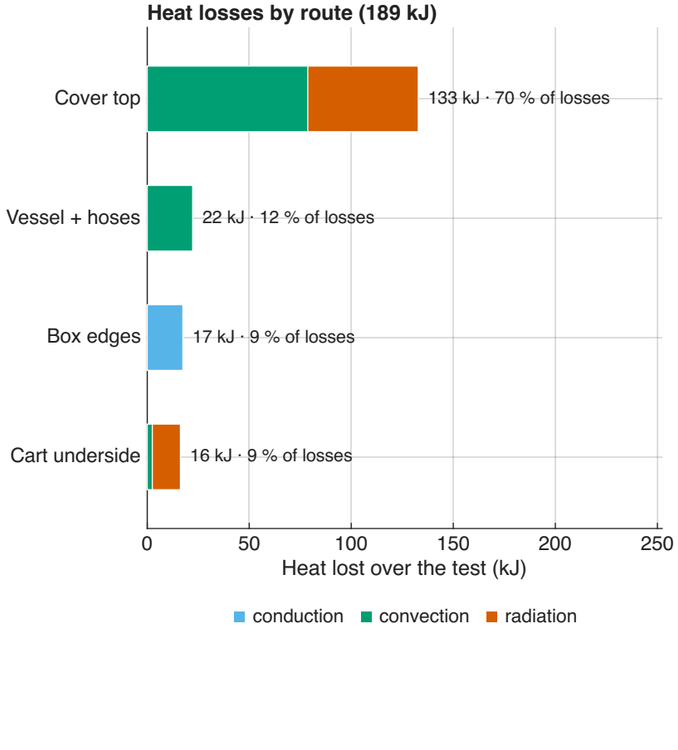

Where the 474 W of lamp light on the cover goes
Power (W)
Time (min)
0
50
100
150
200
250
300
350
400
450
0
5
10
15
20
25
30
Never absorbed
287 kJ · 34 %
Radiation loss
68 kJ · 8 %
Convection loss
103 kJ · 12 %
Conduction loss
17 kJ · 2 %
Warms collector parts
200 kJ · 23 %
Heats the water
177 kJ · 21 %


Energy split, 853 kJ on the cover
Never absorbed
287 kJ · 34 % of the cover light
Into the water
177 kJ · 21 % of the cover light
Kept by parts
200 kJ · 23 % of the cover light
Lost to the room
189 kJ · 22 % of the cover light
conduction 17 · convection 103 · radiation 68 kJ
0
100
200
300
400
500
Energy over the test (kJ)


Heat losses by route (189 kJ)
Cover top
133 kJ · 70 % of losses
Vessel + hoses
22 kJ · 12 % of losses
Box edges
17 kJ · 9 % of losses
Cart underside
16 kJ · 9 % of losses
0
50
100
150
200
250
Heat lost over the test (kJ)
conduction
convection
radiation